

TrES-3b

R-0419

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180620 · created 2026-07-19T08:03:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458289.9477 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.028
Transit depth (Rp/Rs)^2	3.83 +/- 1.09 [%]
Orbital Inclination (inc)	81.19 +/- 0.82
Airmass coefficient 1 (a1)	1.061 +/- 0.011
Airmass coefficient 2 (a2)	-0.051 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [229, 382]
Optimal Aperture	4.65
Optimal Annulus	9.26
Transit Duration (day)	0.0485 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.028 vs 0.16309 (deviation 1.18σ)
Duration fit vs published	0.0485 d vs 0.0567 d (deviation 0.93σ)
Mid-time O–C	-1.8 min
Depth SNR	7.0
Residual scatter	1.07 %

Light curve

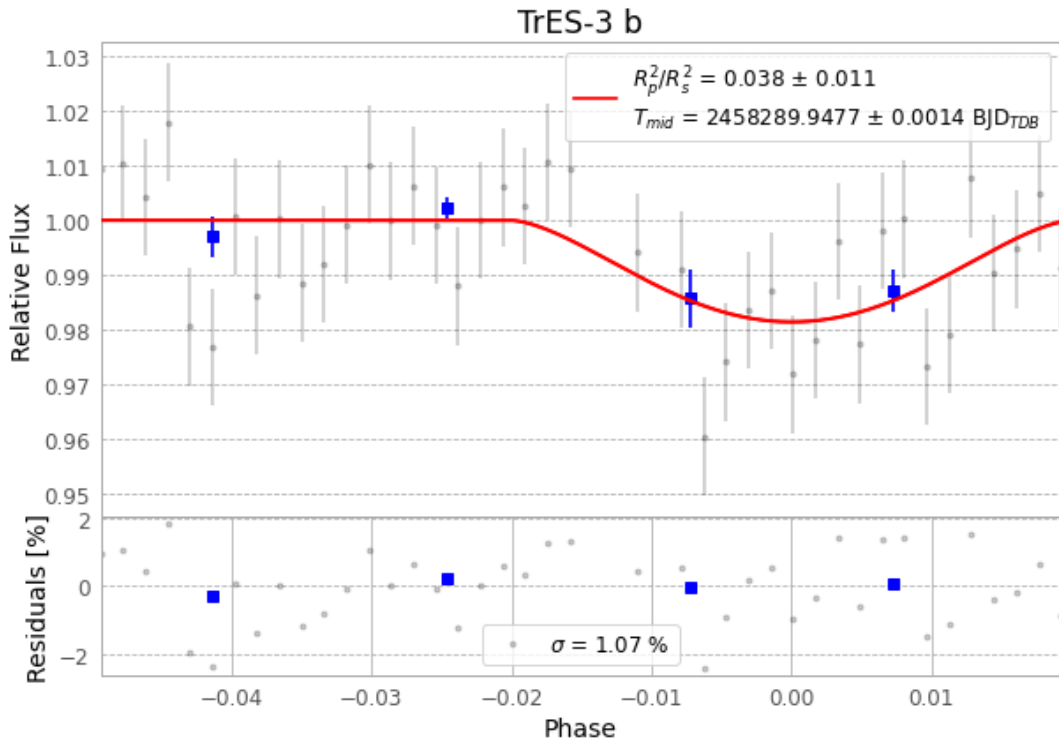


Figure 1 — FinalLightCurve_TrES-3 b_2018-06-20.png (SHA-256 bf668a2faba12ef2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [299, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 347e7c82c7fc9263...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

44 FITS frames (29 MB), TRES-3180620090612.FITS → TRES-3180620111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180620.log (d40acc2e40ad...), FinalLightCurve_TrES-3 b_2018-06-20.png (bf668a2faba1...), FinalLightCurve_TrES-3 b_2018-06-20.csv (454e00e795b1...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 08:03Z.